

Living Literature: A Collaborative VR Experience for Enhancing Shared Reading Through Affective and Generative Feedback

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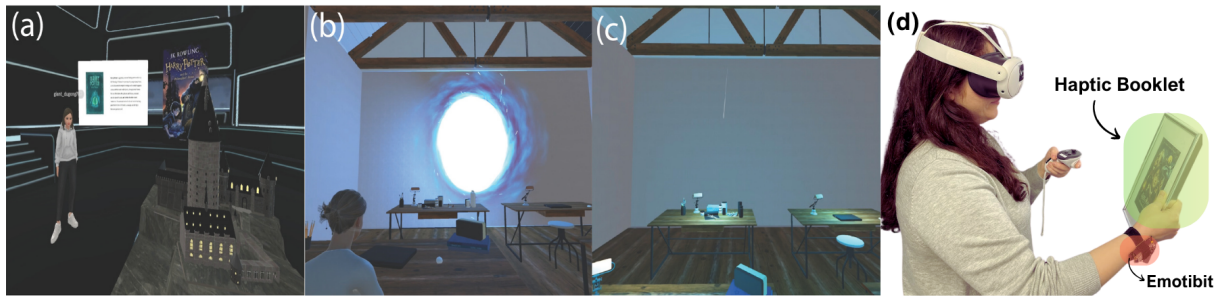


Figure 1: Our system enhances social reading through immersive VR: (a) Two remote readers meet in a shared virtual library to read Harry Potter; a 3-D Hogwarts model and a synchronised e-book page anchor their discussion. (b) When the conversation turns to the story’s “mysterious portal,” the system detects excitement and materialises a shimmering gateway. (c) As collective arousal rises and fearful language emerges, the room’s lighting dims, heightening tension. (d) A user wearing a Meta Quest 3 headset, an EmotiBit PPG sensor, and a handheld “haptic booklet”

Abstract

Traditional e-book reading lacks the shared sensory experiences that deepen engagement. We present a system that transforms virtual reading into a dynamic, multisensory, collaborative experience. Our system integrates Virtual Reality (VR), haptics, generative audio, and physiological sensing to create a shared environment that reacts to the e-book’s narrative and the readers’ emotional state. As users read and discuss a story, the system analyses their conversation sentiment and physiological arousal to dynamically generate

VR visuals, soundscape, and haptic events, providing shared embodied anchors for key narrative moments.

CCS Concepts

• **Human-centered computing** → *Virtual reality; Natural language interfaces; Haptic devices.*

Keywords

Virtual Reality, Generative AI, LLM, Multisensory Experience

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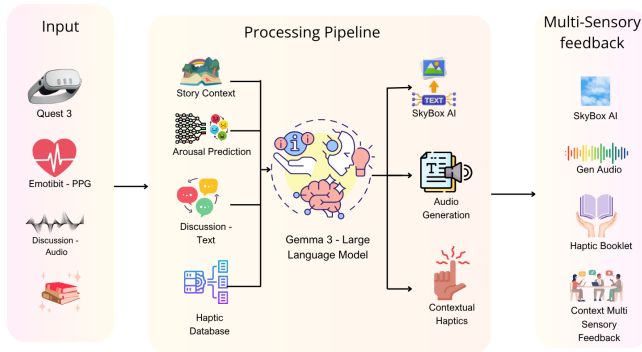


Figure 2: System Architecture for Living Literature

1 Introduction

Reading is a deeply personal and emotional activity, while book clubs make it a shared experience. However, current digital reading platforms (e.g., video calls) don't capture the rich atmosphere of an in-person gathering. They lack the multisensory and emotional context that comes from sharing a physical space, making it hard for readers to feel connected. Adding sensory feedback can enrich storytelling. For example, adding haptics to stories can make them more memorable [Israr et al. 2014], and immersive media with synchronised sensory cues can heighten emotions [Armstrong et al. 2023; Gupta et al. 2024]. This suggests a space to transform digital reading into a multisensory, bio-responsive group experience.

VR can bring readers into a shared virtual space, yet most multi-user VR spaces have static backdrops that do not reflect the narrative's mood or user's feelings. We ask: Can a VR dynamically respond to the story and the readers' emotions in real time? We present Living Literature, a VR space that evolves based on the story, the live group discussion, and physiological signals. The system listens to the conversation (via speech-to-text and sentiment analysis), monitors collective arousal (via PPG sensors), and aligns these with narrative moments to trigger changes in VR visuals, sounds, and haptics.

Our work's novelty lies in integrating these feedback loops to create a shared experience that is both context-aware and personalised to the group's emotional state. Our contributions include: (1) a novel architecture for a responsive multi-user literary VR experience that dynamically blends pre-authored story events with generative audio-visual effects; (2) using live verbal discussion and real-time physiological data to generatively adapt a shared virtual reading environment; and (3) a demonstration of how combining VR, haptics booklet, and AI can deepen engagement, social presence, and emotional resonance in a collaborative reading session.

2 Prototype Implementation

Each participant uses a Meta Quest 3¹ headset and EmotiBit² wrist sensor streaming 100 Hz PPG for heart-rate variability. They hold a Haptic Booklet containing eight LRAs, bone-conduction speaker, and ESP32-S3 microcontroller with Bluetooth connectivity. The

booklet pairs with the host PC, enabling complex vibration patterns and audio cues for tactile-auditory congruence. All devices connect via Wi-Fi 6 to a central server for synchronized multi-user updates. Our Unity environment uses Netcode-for-Entities with Photon relay³ for state consistency across three asynchronous pipelines. The input pipeline ingests XML-tagged narratives, Whisper-large⁴ speech transcripts, and PPG data processed by a 1D-CNN trained on WESAD⁵ for continuous arousal scores. The affective-generative core packages recent transcripts with arousal values, querying local Gemma3-12B⁶ to produce textual prompts for skyboxes, audio prompts for soundscapes, and symbolic haptic profiles. The output pipeline generates 8K HDRI skyboxes via SkyBox AI⁷, refines props using Stable Diffusion XL⁸, creates 12-second stems with Meta MusicGen⁹ cross-faded via FMOD¹⁰, and streams haptic patterns over BLE via OSC¹¹. Server-timestamped events broadcast via UDP with local interpolation enable ten-second updates to skyboxes, lighting, audio, and haptics upon detecting group emotion changes.

3 Formative Study - Social Dynamics in Shared Reading

To understand how our system affects natural conversation flow and group cohesion, we conducted a formative study with 6 participants (aged 22-34, 2 female, 4 male) organised into 2 groups of three. Each group participated in two 15-minute collaborative reading sessions using Chapter 3 of Harry Potter and the Philosopher's Stone: one baseline condition with standard VR reading (no environmental feedback) and one with our full Living Literature system enabled, with conditions counterbalanced across groups. Video analysis revealed that while baseline sessions showed consistent turn-taking patterns ($M = 12.3$ turns per participant, $SD = 2.1$), the system-enabled condition initially disrupted conversation flow in the first 3-4 minutes, with participants pausing mid-sentence during environmental changes ($M = 8.7$ turns, $SD = 3.2$). However, adaptation occurred rapidly, with conversation returning to baseline levels by minute 7 ($M = 11.8$ turns, $SD = 2.4$). Friend groups demonstrated faster adaptation ($M = 5.2$ minutes) compared to stranger groups ($M = 7.8$ minutes), and post-session interviews revealed that 85% of participants found the environmental responses "enhanced the shared experience" despite initial distraction. Notably, friend groups exhibited 23% more gestural communication in VR space when discussing emotionally charged passages, suggesting the system amplified existing social bonds. Stranger groups reported feeling "more connected to the story and each other" (4.2/5 vs 3.1/5 in baseline), indicating the system's potential for facilitating social connection through shared emotional experiences. These findings informed our system's design choices and revealed the importance of gradual introduction of environmental changes, leading us to plan a future study comparing the effectiveness of individual

³Unity Netcode-for-Entities

⁴Whisper-large

⁵WESAD

⁶Gemma 3

⁷Blockade Labs SkyBox AI

⁸Stable Diffusion XL

⁹Meta MusicGen

¹⁰FMOD Studio

¹¹Open Sound Control

¹<https://www.meta.com/quest/quest-3/>

²<https://www.emotibit.com/>

modalities (visual, audio, haptic) to optimise the balance between immersion and social flow.

4 Conclusion

We present Living Literature, a multisensory VR reading system that transforms collective reading and discussion into an interactive, affective experience. Our prototype demonstrates that a VR environment can indeed “listen and respond” in real time, creating a feedback loop that heightens engagement and group cohesion. In future work, we will optimize the communication platform to allow users to better experience the fun of interacting with others in this VR environment. We also plan to conduct longitudinal studies with

different groups of participants to assess sustained engagement and cross-cultural applicability.

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